

# SHAM x prothioconazole: does AOX contribute to the temperature dependence of azole potency?

*Zymoseptoria tritici* IPO323 · PreSens SensorDish 24-vial plates · YMS medium · growth rate from dissolved-oxygen traces (manuscript oxygen model). Design sheet v4, 14 September 2026.

## 1. Question and predictions

Figure 6 shows that the alternative oxidase (AOX) carries respiration at 27 °C (azide) and that blocking it with SHAM costs growth at 27 °C but not at 15 °C. Neither experiment contained prothioconazole, so they show that AOX matters under supra-optimal heat, not that it is part of why prothioconazole loses potency there. This experiment adds the drug at two concentrations, with and without SHAM, at both temperatures. The question is whether SHAM increases the drug's effect at 27 °C and not at 15 °C.

| Outcome                                   | 27 °C                                                                                                                                                                                | 15 °C                                                                  | Reading                                                                                           |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| AOX contributes to the temperature effect | Drug + SHAM inhibits growth <b>more</b> than the multiplicative prediction from the two single agents, at both drug doses (AOX was blunting the drug; blocking it restores potency). | SHAM alone has no effect; drug + SHAM equals drug alone at both doses. | Title: AOX <i>contributes to</i> the temperature dependence of potency.                           |
| AOX matters but independently of the drug | Drug + SHAM equals the multiplicative prediction (the two effects are independent).                                                                                                  | As above.                                                              | Association paper; title stays soft. Still publishable.                                           |
| General drug interaction                  | Potentialiation present.                                                                                                                                                             | Potentialiation of the same size.                                      | SHAM x drug interaction that is not temperature-specific; does not support the temperature claim. |

Primary statistic, per biological replicate, temperature and drug dose  $d$ : the multiplicative interaction contrast  $I_d = \log r_{PS,d} - \log r_{P,d} - \log r_{S,d} + \log r_V$ , where  $r$  is the mean growth rate of the condition's wells (V vehicle, S SHAM alone, P drug alone, PS drug + SHAM).  $I = 0$  means the two agents act independently on log growth rate;  $I < 0$  means SHAM potentiates the drug. The temperature question is  $I_{27} - I_{15}$ , computed within each replicate (both plates of a replicate share one master), at each dose and pooled over the two doses. Expected if AOX contributes:  $I_{27} < 0$  at both doses,  $I_{15} \sim 0$ , and the difference has the same sign in all three cultures. Primary contrast at 27 °C is the 2 mg/L dose (its drug + SHAM well stays measurable even under strong potentialiation); 4 mg/L is secondary. At 15 °C both doses are primary.

## 2. Design at a glance

| Item                  | Choice                                                                                                                                                                                                                                         | Why                                                                                                                                                                                                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Temperatures          | 15 and 27 °C, one plate each, from the same inoculated master on the same day                                                                                                                                                                  | Same contrast as Fig. 6; 15 °C is the negative control for off-target effects                                                                                                                                                                    |
| Conditions per plate  | Prothioconazole 0, 2, 4 mg/L x SHAM 0 / 0.2 mM = 6 conditions. Vehicle 5 wells, SHAM alone 5, the four drug conditions 3 each, plus 2 cell-free blanks (vehicle, SHAM)                                                                         | 3 x 2 factorial, identical at both temperatures, so dose is not confounded with temperature. Extra wells go to the two no-drug conditions because they are the denominators of every contrast. The SHAM blank catches abiotic O <sub>2</sub> use |
| Prothioconazole doses | <b>2 and 4 mg/L on every plate.</b> Measured relative growth in the Fig. 2 dose series: 2 mg/L gives 0.36 at 15 °C and 0.80 at 27 °C; 4 mg/L gives 0.26 and 0.61. (1 mg/L was dropped: it gives 0.99 at 27 °C, i.e. no drug effect to modify.) | Both doses are partial at both temperatures. 2 mg/L is the primary contrast at 27 °C because its                                                                                                                                                 |

| Item       | Choice                                                                                                                                                                | Why                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Plates     | 6 (3 replicates x 2 temperatures) + 1 washout plate (Section 12)                                                                                                      |                                           |
| Incubators | Verified flat beforehand (Section 3); use the same two incubators for all three replicates, 27 °C plate never in the incubator that produced the Fig. S16 oscillation | Removes the periodic artefact Gab flagged |
| Run length | 27 °C: ~70 h · 15 °C: 72-96 h, dark                                                                                                                                   | As the SHAM series (68.6 h)               |

### 3. Before the first plate: incubator check (1 night, no cells)

Fill a plate with medium only (5.05 mL per vial, 0.4% DMSO, 0.2% ethanol), put it in each candidate incubator at its working temperature and log overnight. Accept an incubator only if the trace, after removing the slow trend, stays within about 0.05 mg/L and shows no cycle of 1-4 h. The incubator used for the Fig. 6a 27 °C plate shows a 1.5-4 h cycle growing to 1.5 mg/L peak-to-peak; do not use it for the 27 °C plates. Send me the export and I will run the same residual/period check as script 44.

### 4. Stocks

| Stock             | Make                                                                              | Per vial | Final                | Notes                                                                                                                                                                                      |
|-------------------|-----------------------------------------------------------------------------------|----------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P4                | 1 mg/mL prothioconazole in DMSO (your standard stock, neat)                       | 20 µL    | 4 mg/L, 0.4% DMSO    | PESTANAL standard; fresh dilution each set-up day                                                                                                                                          |
| P2                | 0.5 mg/mL: 500 µL of 1 mg/mL + 500 µL DMSO                                        | 20 µL    | 2 mg/L, 0.4% DMSO    |                                                                                                                                                                                            |
| P0 (no drug)      | neat DMSO                                                                         | 20 µL    | 0.4% DMSO            | Every no-drug vial gets this                                                                                                                                                               |
| SHAM 100 mM       | 250 mM master in absolute ethanol (76.6 mg in 2 mL), then 400 µL + 600 µL ethanol | 10 µL    | 0.2 mM, 0.2% ethanol | Foil-wrapped, made fresh from the master. <b>Before set-up day</b> , put 1 mL of the 100 mM working stock and a 0.2 mM dilution in YMS at 15 °C overnight and confirm nothing precipitates |
| Ethanol (vehicle) | absolute ethanol                                                                  | 10 µL    | 0.2% ethanol         | Every no-SHAM vial gets this                                                                                                                                                               |

Golden rule: every vial receives 20 µL from the drug channel (P0, P2 or P4); the inhibitor channel (ethanol or SHAM, 0.2% v/v) is already in the cup mix. Total solvent is identical in all 24 vials of every plate. Per-vial SHAM/ethanol equivalent: 10 µL in 5.05 mL.

### 5. Cells

Three precultures from frozen stock into liquid YMS, 4 days at 21 °C, 200 rpm, one preculture per replicate started on a different day (this is what makes them independent). On the set-up day pass through the 100 µm strainer, measure OD<sub>600</sub>, and inoculate each arm cup to OD<sub>600</sub> = 0.0005 as in the main assay: µL of preculture = 0.0005 x 150 000 ÷ measured OD = 75 ÷ OD. Both arms receive the identical volume from the same preculture. Also take a haemocytometer count for N<sub>0</sub> as usual.

### 6. Per replicate: cups (feeds both plates)

Every cup fits a 250 mL sterile cup and carries a generous excess over what the vials take. There is no separate master: each arm is inoculated directly from the same preculture at the same ratio, which keeps the two arms at the same cell density within pip

| Cup            | Recipe                                              | Vials it fills                            | Needed | Make         |
|----------------|-----------------------------------------------------|-------------------------------------------|--------|--------------|
| Blank, vehicle | 24.95 mL sterile YMS + 50 µL absolute ethanol       | bV: 1 per plate, 2 in total, 5.03 mL each | 10 mL  | <b>25 mL</b> |
| Blank, SHAM    | 24.95 mL sterile YMS + 50 µL 100 mM SHAM (= 0.2 mM) | bS: 1 per plate, 2 in total, 5.03 mL each | 10 mL  | <b>25 mL</b> |

Then, per vial: **5.03 mL of the cup mix + 20 µL of the drug stock** the map calls for (P0, P2 or P4) = 5.05 mL. DMSO 0.40%, ethanol 0.20% in every vial. Fill both plates from the same cups, seal, and start both readers within the hour. YMS needed per replicate: 300 mL arms + 50 mL blanks = 350 mL; have 400 mL sterile.

## 7. Plate map (identical for the 15 °C and 27 °C plate and for all three replicates)

|   | 1   | 2   | 3   | 4   | 5   | 6   |
|---|-----|-----|-----|-----|-----|-----|
| A | V   | V   | V   | V   | V   | bV  |
| B | P2  | P2  | P2  | P4  | P4  | P4  |
| C | S   | S   | S   | S   | S   | bS  |
| D | P2S | P2S | P2S | P4S | P4S | P4S |

| Code | Cup mix (5.03 mL)   | Drug stock added (20 µL) | Inhibitor (already in the cup, nothing added per vial) | Wells | n | Role                               |
|------|---------------------|--------------------------|--------------------------------------------------------|-------|---|------------------------------------|
| V    | Vehicle arm (cells) | P0                       | ethanol                                                | A1-A5 | 5 | Solvent control (denominator)      |
| S    | SHAM arm (cells)    | P0                       | SHAM 0.2 mM                                            | C1-C5 | 5 | SHAM alone (denominator)           |
| P2   | Vehicle arm (cells) | P2                       | ethanol                                                | B1-B3 | 3 | Drug 2 mg/L                        |
| P4   | Vehicle arm (cells) | P4                       | ethanol                                                | B4-B6 | 3 | Drug 4 mg/L                        |
| P2S  | SHAM arm (cells)    | P2                       | SHAM 0.2 mM                                            | D1-D3 | 3 | Drug 2 mg/L + SHAM                 |
| P4S  | SHAM arm (cells)    | P4                       | SHAM 0.2 mM                                            | D4-D6 | 3 | Drug 4 mg/L + SHAM                 |
| bV   | Blank, vehicle      | P0                       | ethanol                                                | A6    | 1 | Cell-free control                  |
| bS   | Blank, SHAM         | P0                       | SHAM 0.2 mM                                            | C6    | 1 | Abiotic O <sub>2</sub> use by SHAM |

Pipetting logic: rows A and B are filled from the vehicle arm, rows C and D from the SHAM arm; columns 1-3 of rows B and D get P2, columns 4-6 get P4; rows A and C get P0; the two column-6 vials of rows A and C are the cell-free blanks. Condition is confounded with row, but the primary contrast  $I = \log r_{P5} - \log r_P - \log r_S + \log r_V$  spans all four rows with weights +1, -1, -1, +1, so any linear row gradient cancels exactly in I. The same map is used on all six plates.

## 8. Running the plates

| Step               | Detail                                                                                                                                                                                                     |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Export             | PreSens export as usual, sheet 'Oxygen Original'. File names: data/aox/ptc_sham_rep{1,2,3}_{15,27}_Oxygen.xlsx. One layout file, data/aox/ptc_sham_layout.csv (well, condition), shared by all six plates. |
| Between replicates | Nothing carries over: fresh preculture, fresh drug and SHAM dilutions, same two incubators (15 °C and 27 °C each in their own, as in the SHAM series), same plate map.                                     |

## 9. Timeline (two verified incubators)

| Day   | Action                                                                                                                                            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| -6    | Start precultures R1 and W. Run the incubator check plates overnight (Section 3).                                                                 |
| -4    | Start preculture R2.                                                                                                                              |
| -2    | Start preculture R3. Washout: expose W to SHAM for 24 h (Section 12).                                                                             |
| -1    | Washout plate on at 27 °C (Section 12).                                                                                                           |
| 0     | Set up R1: master, arms, both plates; start readers. (~3 h bench)                                                                                 |
| 2     | Washout plate off (export). Set up R2 if two further incubators are free; otherwise day 4 after R1 comes off.                                     |
| 4     | R1 27 °C plate off (export). 15 °C plate may run to day 4-5.                                                                                      |
| 4-5   | Set up R2 (both plates).                                                                                                                          |
| 8-9   | R2 off; set up R3.                                                                                                                                |
| 12-13 | R3 off. All six exports in the repo; fit windows in the trim selector (script 03); analysis scripts 45 (rates) and 57 (figure) run on the tables. |
| ~15   | Result: interaction index at each temperature, figure in the Fig. 6 style, one paragraph for Results.                                             |

With one incubator per temperature and precultures staggered as above, the wet work is about 13 days; with only one incubator in total, about four weeks.

## 10. Analysis plan (fixed before the data arrive)

1. Fit every trace with the manuscript oxygen model  $O(t) = O_0 + (K/r)(1 - e^{rt})$  on hand-selected intervals from the trim selector, exactly as for Fig. 6a; growth rate  $r$  per well. The two cell-free blanks are inspected for abiotic drawdown (SHAM) and reported, not subtracted, as in the SHAM series.
2. Per plate, mean  $r$  per condition (V, S, P2, P4, P2S, P4S) and also  $K$ , the oxygen-consumption parameter.  $K$  scales with per-cell respiration  $\times N_0$ , so a treatment that kills cells lowers  $K$  and leaves  $r$  alone, whereas inhibition lowers  $r$  and leaves  $K$ ; in the Fig. 6a data SHAM at 27 °C lowered  $r$  to 0.35-0.38 of control with  $K$  unchanged. The same  $r/K$  comparison is reported here.
3. Interaction contrast  $I_d$  per plate and dose (Section 1), and its temperature difference  $I_{27,d} - I_{15,d}$  within each replicate. Primary: that difference at 2 mg/L, across the three replicates, with a 95% CI and the sign in each culture; paired  $t$  as the formal test, with the caveat that three pairs cannot give a permutation  $P$  below 0.125. Secondary: 4 mg/L; the two doses pooled; one-sample tests of  $I$  at each temperature. A drug + SHAM well whose fit returns  $r$  at the model's lower bound is treated as censored ( $r$  below the identifiability floor), which counts as potentiation in direction but is not used as a number.
4. The same quantities as a linear model on  $\log r$  with temperature  $\times$  dose  $\times$  SHAM, culture as a blocking factor; this estimates the same contrasts and is reported as the check that the two agree.
5. Potency framing for the text: drug effect without SHAM (P/V) and with SHAM (PS/S) at each dose and temperature. The prediction is  $PS/S < P/V$  at 27 °C and  $PS/S \sim P/V$  at 15 °C.
6. Robustness: the seven automated window rules of Fig.

Expected floor. Under independence the 4 mg/L + SHAM wells at 27 °C sit near  $0.61 \times 0.38 = 0.23$  of control,  $r$  about  $0.009 \text{ h}^{-1}$ , close to where the 0.35 mM SHAM wells became non-identifiable; the 2 mg/L + SHAM wells sit near 0.30 of control,  $r$  about 0.012, with room to fall. That is why 2 mg/L is the primary 27 °C contrast. If the 2 mg/L + SHAM wells also hit the floor in R1, the effect is large and real but unquantifiable: drop SHAM to 0.1 mM for R2 and R3 and report R1 as censored.

## 12. Washout plate (Nick's question: does SHAM inhibit or kill?)

One extra plate, 27 °C only, no prothioconazole. Cells are exposed to SHAM for 24 h at high density, the SHAM is removed by dilution (no centrifugation), and the plate asks whether growth comes back. If cells exposed to 0.2 mM and 1 mM SHAM regrow like vehicle-exposed cells, SHAM is a reversible inhibitor and the Figure 6 reading stands; if they stay flat, it was toxic. This uses its own preculture (W) and runs in the 27 °C incubator alongside R1.

| Step                        | Detail                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day -6                      | Start preculture W (together with R1), 4 days at 21 °C, 200 rpm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Day -2: exposure tubes      | Strain, measure $OD_{600}$ . Make 40 mL YMS at $OD_{600}$ 0.05 (100x the plate density; $\mu\text{L}$ of preculture = $2000 \div OD$ ), no solvent yet. Split into 9 x 4 mL tubes and add per tube: vehicle, 16 $\mu\text{L}$ ethanol + 16 $\mu\text{L}$ DMSO; 0.2 mM, 8 $\mu\text{L}$ 100 mM SHAM + 8 $\mu\text{L}$ ethanol + 16 $\mu\text{L}$ DMSO; 1 mM, 16 $\mu\text{L}$ 250 mM SHAM + 16 $\mu\text{L}$ DMSO. Every tube is then 0.4% ethanol, 0.4% DMSO. Three tubes per condition. 24 h at 27 °C, 200 rpm, dark. The tubes are technical triplicates from one preculture; say so. |
| Day -1: washout by dilution | Measure $OD_{600}$ of each tube. Dilute each tube into fresh YMS to $OD_{600}$ 0.0005 ( $\mu\text{L} = 0.0005 \times 15000 \div OD$ ) into 15 mL; always at least 1:100, so residual SHAM is at most 0.002 mM from the 0.2 mM tubes and 0.01 mM from the 1 mM tubes, below the lowest dose in Fig. 6a). Each 15 mL: 14.97 mL YMS + 30 $\mu\text{L}$ ethanol + cells. Fresh never-exposed control: 8 mL YMS + 16 $\mu\text{L}$ ethanol + preculture W to $OD$ 0.0005.                                                                                                                    |
| Day -1: plate               | 5.03 mL of each dilution + 20 $\mu\text{L}$ DMSO (P0) per vial = 5.05 mL, 0.4% DMSO, 0.2% ethanol. Map below. 27 °C, dark, ~70 h.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Day +2                      | Export as data/aox/sham_washout_Oxygen.xlsx; layout data/aox/sham_washout_layout.csv.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

After the dilution to  $OD$  0.0005 (at least 1:100) the exposure solvents are negligible and the plate is at the standard 0.2% ethanol, 0.4% DMSO.

|   | 1   | 2   | 3     | 4     | 5   | 6   |
|---|-----|-----|-------|-------|-----|-----|
| A | W-V | W-V | W-0.2 | W-0.2 | W-1 | W-1 |
| B | W-V | W-V |       |       |     |     |

### 13. Reagent and consumable checklist per replicate

| Item                                         | Amount                                                         |
|----------------------------------------------|----------------------------------------------------------------|
| YMS medium, sterile                          | 400 mL (300 arms + 50 blanks + spare)                          |
| Prothioconazole 1 mg/mL in DMSO              | 1 mL (P4 neat, 0.12 mL used; P2 made from 500 µL of it)        |
| DMSO                                         | 2 mL (0.5 mL for P2, 0.5 mL as P0, spare)                      |
| SHAM 250 mM in ethanol (foil)                | 0.4 mL master → 1 mL of 100 mM working stock (0.35 mL used)    |
| Absolute ethanol                             | 2 mL (0.6 mL for the SHAM dilution, 0.35 mL as vehicle, spare) |
| SensorDish plates (24 vials)                 | 2, with lids/seals (plus 1 for the washout plate, once)        |
| Sterile 250 mL cups                          | 4 (two arms, two blanks)                                       |
| Preculture, 4 days, 21 °C                    | 1 flask (independent per replicate)                            |
| 100 µm strainer, haemocytometer, OD cuvettes | as usual                                                       |